Frank Bieder

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Research Scientist and Group Leader in visual and spatial Learning for autonomous systems. My work centers on scalable training data generation and cross-sensor domain adaptation - bridging HD maps, LiDAR, and camera perception to train robust real-world models under limited annotation. I have led research groups, principal-investigated six-figure industrial grants, published across NeurIPS, CVPR, IROS, RAL, T-ITS and T-IV, and deployed perception systems on autonomous research vehicles across multiple European Cities.

Education

May 2020 – Oct 2025 **Karlsruhe Institute for Technology**, **PhD/Dr.-Ing.** – Autonomous Systems / Deep Learning
Thesis topic: Learning from Maps: Scalable Ground Truth Generation in Autonomous Driving.
Supervisor: Christoph Stiller. Defense: Mar 26. Summa cum Laude, 1.0/1.0 with Distinction.

Aug. 2022 – Oct 2022 **UC Berkeley – BAIR/Mechanical Systems Control Lab, Visiting PhD Candidate** - Physical AI
Research on map-less driving and map perception. Supervisor: Masayoshi Tomizuka.

Oct. 2014 – Aug 2018 **Karlsruhe Institute of Technology, M.Sc.** – Electrical Engineering / Computer Science
GPA: 1.4/1.0 - Focus: Machine learning, system theory & robotics. Master's thesis: Supervised Domain Adaptation for Semantic Segmentation. Grade: 1.0/1.0.

Aug. 2015 – May 2016 **University of Ottawa, Exchange Studies M.Sc.** – Electrical Engineering / Computer Science
Canadian GPA: 3.8/4.0 - First semester: Lectures focusing on machine learning and data processing. Second semester: Research projects in two university labs.

Oct. 2011 – Oct. 2014 **Karlsruhe Institute of Technology, B.Sc.** – Electrical Engineering / Computer Science
Bachelor's thesis: Fusion of Data from Stereo Camera and Radar Sensor for Tracking of Extended Objects, Grade: 1.0/1.0.

Work Experience

Nov 2025 – today **FZI Research Center for Information Technology – Group Lead for Visual & Spatial Learning**
Leading a team of researchers and PhD students across visual and spatial learning, with a focus on auto-labeling pipelines for multimodal perception and E2E-driving systems.
E2E-Owner and Co-adviser for the KITScenes Multimodal dataset.

May 2019 – Oct 2025 **FZI Research Center for Information Technology – Research Scientist and PhD Candidate**
Supervisor: Christoph Stiller. Specializing in map perception and cross-sensor domain adaptation. Project led on multiple six-figure (industrial) research projects. Co-maintenance of software and hardware stacks for (up to) two AD research vehicles.

Oct. 2018 - April 2019 **Atlatec GmbH, Karlsruhe (AI/AD start-up, acquired by Bosch in 2022) – Deep Learning Lead**
Supervisor: Julius Ziegler. Developing of multimodal CNNs & weakly supervised learning methods for the automatic generation of high precision planning maps.

Nov. 2017 – Aug. 2018 **Daimler AG Image Understanding Group, Stuttgart – Master Thesis**
Supervised by Uwe Franke / Fiete Botschen. Worked on domain adaptation: Use of synthetic data to improve the pixel-wise semantic classification of images in real-world applications.

May 2016 – Oct. 2017 **FZI Research Center for Information Technology, Karlsruhe – Research Assistant**
Supervisor: Sascha Wirges. Research projects in environment perception / machine learning.

Jan. 2016 – May 2016 **University of Ottawa VIVA Lab, Ottawa Canada – Research Assistant**
Supervisor: Robert Laganiere. Research project in RGB-D instance segm. of indoor scenes.

Jan. 2015 – Jul. 2015 **Mercedes-Benz Research & Development, Sunnyvale USA – Research Internship**
Supervisor: Jörg Hillenbrand. Research internship in MBRDNA's Vehicle Intelligence Group.
Developed a framework to evaluate the driving behavior of an autonomous vehicle.

Publications (selection)

Frank Bieder, Hendrik Königshof, Haohao Hu, Fabian Immel, Yinzhe Shen, Jan-Hendrik Pauls, Christoph Stiller. XD-MAP: Cross-Modal Domain Adaptation using Semantic Parametric Mapping**.** *In CVPR – NexD Workshop*, 2026**.**

Richard Schwarzkopf et al. The Road Ahead in Autonomous Driving: The KITScenes Multimodal Dataset**.** *Arxiv: 2606.02956,* 2026**.** Visit kitscenes.com/multimodal

Fabian Immel, Jan-Hendrik Pauls, Richard Fehler, Frank Bieder, Jonas Merkert, Christoph Stiller. SDTagNet: Leveraging Text-Annotated Navigation Maps for Online HD Map Construction**.** *In NeurIPS,* 2025.

Fabian Immel, Richard Fehler, Frank Bieder, Jan-Hendrik Pauls, Christoph Stiller. M3TR: A Generalist Model for Real-World HD Map Completion**.** *IEEE/RAS Robotics and Automation Letters (RAL)*, 2025.

Frank Bieder, Haohao Hu, Johannes Schantz, Oguzahn Kirik, Florian Ries, Martin Haueis, Christoph Stiller. Map Learning: Ein Ansatz zur automatisierten Erstellung von Trainingsdaten unter Verwendung von HD-Karten und Mehrfachbefahrungen. *15. FAS Workshop,* 2023. **1nd Best Paper Award**

Sven Richter, Frank Bieder, Sascha Wirges, Christoph Stiller. A Dual Evidential Top-View Representation to Model the Semantic Environment of Automated Vehicles. *In IEEE Transactions on Intelligent Vehicles (T-IV),* 2024.

Haohao Hu, F. Han, Frank Bieder, Jan-Hendrik Pauls, Christoph Stiller. TEScalib: Targetless Extrinsic Self-Calibration of LiDAR and Stereo Camera for Automated Driving Vehicles with Uncertainty Analysis. *IEEE/RSJ IROS*, 2022.

Christoph Burger, Johannes Fischer, Frank Bieder, Ömer Tas, Christoph Stiller. Interaction-Aware Game-Theoretic Motion Planning for Automated Vehicles using Bi-level Optimization. *IEEE ITSC,* 2022. **2nd Best Paper Award** .

Kunyu Peng, Juncong Fei, Kailun Yang, Alina Roitberg, Jiaming Zhang, Frank Bieder, Philipp Heidenreich, Christoph Stiller, Rainer Stiefelhagen. MASS: Multi-Attentional Semantic Segmentation of LiDAR Data for Dense Top-View Understanding. In *IEEE T-ITS,* 2022.

Frank Bieder, Maximilian Link, Simon Romanski, Haohao Hu, Christoph Stiller. Improving Lidar-Based Semantic Segmentation of Top-View Grid Maps by Learning Features in Complementary Representations. *In FUSION,* 2021.

Scholarships/Awards

Nov 2023	**1st Best Paper Award Uni-DAS FAS Workshop** – First author of paper
Oct 2022	**2nd Best Paper Award IEEE ITSC 2022** – Co-author of paper
Aug 2022 – Oct 2022	**KHYS Exchange Grant** – Funding three months of study exchange with UC Berkeley
Nov 2021	**FUSION Travel Grant** – Conference fees, travel costs & accommodation in South Africa
Nov 2019 – today	**KSOP Scholarship** – Fully funding research of one PhD year and an MBA program in parallel
Jun 2016 – Jun 2019	**Daimler Student Partnership** – Scholarship covering personal mentoring and trainings
Sep 2015 – Dec 2015	**Ontario BaWü Program** – Fully funding one academic exchange year in Canada
Aug 2015 – May 2016	**German Academic Exchange Service Promos** – Scholarship for international studies
Jul 2011	**GDCH Award** – German Chemical Society's award for excellence in the high school diploma

Skills and Engagements

Languages	German (native), English (fluent), French (upper intermediate)
Technical Skills	Python (Pytorch, TensorFlow, OpenCV, numpy), C++, ROS, containerized workflows
Supervision	Supervised 11 MSc thesis projects, currently co-supervising 7 PhD students
Scientific Workshops	1st Organizer: FMAD Foundation Models for AD (ITSC 24/25), MB2ML Mapless AD (IV 22/23)
MBA Coursework	Completed MBA Fundamentals Program, Hector School of Engineering and Management
Voluntary Work	Engineers Without Borders (2012–2013); International Student Mentoring (2019–2022)
Further Interests	Hiking, sailing, scuba diving